

COST FUNCTION IN REGRESSION + ERROR METRICS

Topic

- Cost Function in Regression
 - Residual / Error
 - MSE, RMSE, MAE
 - Why model minimizes cost
 - Model error comparison (practical)
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1. Learning Objectives

By the end of this session, students will be able to:

- Understand what **error/residual** is
 - Explain what a **cost function** is
 - Understand **MSE, RMSE, MAE** clearly
 - Know why ML models try to **minimize cost**
 - Compare models using error metrics
 - Interpret which model is better and why
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2. Quick Recap (Connect with Previous Session)

Start class like this:

“In the previous session, we built Linear and Polynomial Regression models.
Now the question is — how do we know which model is better?
For that, we need to measure how much error our model is making.”

3. Core Concept: What is Error?

3.1 Definition

Error = Actual Value – Predicted Value

$$Error = y_{actual} - y_{predicted}$$

3.2 Real-Life Example

Suppose:

Monthly Salary

Error:

$$Error = 50000 - 48000 = 2000$$

 This means:

3.3 Important Insight

Error tells us:

- How far prediction is from reality
 - Whether model is accurate or not
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3.4 Residual vs Error

👉 In regression, we often use the term **Residual**

Definition:

Residual = Actual – Predicted

👉 So practically:

- Error = Residual (same concept)
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3.5 Types of Errors

Case 1: Positive Error

- Actual > Predicted
- Model is underestimating

Case 2: Negative Error

- Actual < Predicted
 - Model is overestimating
-

4. **!** Problem with Raw Error

Key Question:

“Can we just add all errors to measure model performance?”

Example:

Total = 0 ❌

👉 But actual error is not zero!

Conclusion:

Positive and negative errors cancel each other

👉 So we need a better way to measure error.

5. 📌 Solution: Cost Function

5.1 Definition

A **Cost Function** is a function that measures the **total error of a model**.

5.2 Simple Meaning

“Cost Function tells us how bad our model is.”

- High cost → bad model
 - Low cost → good model
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5.3 Goal of Machine Learning

“The goal of any ML model is to minimize the cost function.”

5.4 Real-Life Analogy

Example: Exam Marks

- Suppose teacher gives penalty for wrong answers
- Student tries to **minimize mistakes**

👉 Same in ML:

- Model tries to **minimize prediction error**
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6. 📊 Common Error Metrics (Cost Functions)

We mainly use 3:

1. **MAE (Mean Absolute Error)**
 2. **MSE (Mean Squared Error)**
 3. **RMSE (Root Mean Squared Error)**
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7. 📌 MAE – Mean Absolute Error

Definition

$$MAE = \frac{1}{n} \sum |y_{actual} - y_{predicted}|$$

Simple Meaning

Take absolute value of errors and find average.

Example

Errors:

- 1000
- -2000
- 3000

Absolute:

- 1000
- 2000
- 3000

Average:

$$(1000 + 2000 + 3000)/3 = 2000$$

Key Points

- No negative values
 - Easy to understand
 - Less sensitive to outliers
-

Teaching Line

“MAE tells the average error in real units.”

8. 📌 MSE – Mean Squared Error

Definition

$$MSE = \frac{1}{n} \sum (y_{actual} - y_{predicted})^2$$

Simple Meaning

Square the errors, then take average.

Example

Errors:

- $1000 \rightarrow 1000^2 = 1,000,000$
- $-2000 \rightarrow 4,000,000$
- $3000 \rightarrow 9,000,000$

Average:

$$(1M + 4M + 9M)/3 = 4.67M$$

Key Points

- Removes negative sign
- Penalizes large errors more
- Sensitive to outliers

Teaching Line

“MSE punishes large mistakes more strongly.”

9. 📌 RMSE – Root Mean Squared Error

Definition

$$RMSE = \sqrt{MSE}$$

Simple Meaning

Square root of MSE → brings error back to original unit.

Why Needed?

- MSE unit becomes squared (e.g., salary² ❌)
 - RMSE converts it back to normal unit (salary ✔)
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Teaching Line

“RMSE is MSE but in a more interpretable form.”

10. 📊 Comparison of MAE vs MSE vs RMSE

to Outliers

error

error

1SE

11. 🎯 When to Use What?

Use MAE when:

- You want simple interpretation
- Outliers are not important

Use MSE / RMSE when:

- Large errors are very bad
- You want to penalize big mistakes

12. 📌 Why Model Minimizes Cost Function?

Core Idea

ML models adjust parameters (like slope/intercept) to reduce error.

Example

Suppose:

- Line is far from data → high error
- Line moves closer → error reduces

👉 Model keeps adjusting until:

“Error becomes minimum”

Teaching Line

“The model learns by trying different values and selecting the one with the lowest error.”

13. 📊 Visualization Idea (Explain in Class)

Imagine:

- Graph with points
- Line is far → high error
- Line moves closer → low error

👉 This is called:

“**Optimization process**”

14. 🔥 Important Interview Concepts

Q1:

What is a cost function?

👉 A function that measures total error of a model.

Q2:

Why do we square errors in MSE?

👉 To avoid cancellation and penalize large errors.

Q3:

Difference between MAE and RMSE?

👉 MAE is simple average error, RMSE penalizes large errors more.

Q4:

Why do models minimize cost?



To improve prediction accuracy.

15. PRACTICAL (Your Format)



Objective

Compare:

- Linear Regression vs Polynomial Regression using:
 - MAE
 - MSE
 - RMSE
-

```
# =====
# COST FUNCTION IN REGRESSION
# ERROR METRICS: MAE, MSE, RMSE
# REAL-LIFE EXAMPLE: Experience vs Salary
# =====

# Step 1: Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_absolute_error, mean_squared_error

# Step 2: Create dataset
data = {
    "Experience": [1,2,3,4,5,6,7,8,9,10],
    "Salary": [25000,28000,32000,39000,48000,60000,75000,93000,114000,138000]
}

df = pd.DataFrame(data)

print("===== DATASET =====")
print(df)

# Step 3: Define X and y
X = df[["Experience"]]
y = df["Salary"]

# =====
# LINEAR REGRESSION
# =====

# Step 4: Train Linear Model
linear_model = LinearRegression()
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linear_model.fit(X, y)

# Step 5: Predictions
y_pred_linear = linear_model.predict(X)

# =====
# POLYNOMIAL REGRESSION
# =====

# Step 6: Create polynomial features
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

# Step 7: Train Polynomial Model
poly_model = LinearRegression()
poly_model.fit(X_poly, y)

# Step 8: Predictions
y_pred_poly = poly_model.predict(X_poly)

# =====
# ERROR METRICS CALCULATION
# =====

# Step 9: Linear Regression Errors
mae_linear = mean_absolute_error(y, y_pred_linear)
mse_linear = mean_squared_error(y, y_pred_linear)
rmse_linear = np.sqrt(mse_linear)

# Step 10: Polynomial Regression Errors
mae_poly = mean_absolute_error(y, y_pred_poly)
mse_poly = mean_squared_error(y, y_pred_poly)
rmse_poly = np.sqrt(mse_poly)

# =====
# PRINT RESULTS
# =====

print("\n===== LINEAR REGRESSION ERRORS =====")
print("MAE :", round(mae_linear, 2))
print("MSE :", round(mse_linear, 2))
print("RMSE:", round(rmse_linear, 2))

print("\n===== POLYNOMIAL REGRESSION ERRORS =====")
print("MAE :", round(mae_poly, 2))
print("MSE :", round(mse_poly, 2))
print("RMSE:", round(rmse_poly, 2))

# =====
# VISUALIZATION
# =====

plt.scatter(X, y, label="Actual Data", color="blue")

plt.plot(X, y_pred_linear, label="Linear Regression", color="red")
plt.plot(X, y_pred_poly, label="Polynomial Regression", color="green")

```



```
plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Model Comparison")
plt.legend()
plt.show()
```

16. Output Interpretation (Very Important)

After running code, tell students:

Observation:

- Polynomial Regression errors are **lower**
- Linear Regression errors are **higher**



Conclusion:

Polynomial model fits data better

17. Practice Task

- Change polynomial degree to 3
 - Compare errors again
 - Which model is best?
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18. Homework

- Calculate MAE manually for 3 rows
 - Explain difference between MSE and RMSE
 - Write 3 lines: "Why model minimizes cost?"
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19. Final

"In machine learning, building a model is not enough.

We must evaluate how good the model is using error metrics.

A good model is the one that minimizes error and makes accurate predictions."